

Class 9 Science – Work and Energy (Annual Exam 2026)

This document contains **exam-ready answers** strictly as per **CBSE Class 9 NCERT**, followed by **keywords** for scoring full marks.

1. Definition of Work

Answer: In science, work is said to be done when a force acting on an object produces a displacement in the direction of the force. If a force is applied but no displacement occurs, no work is done.

Essential conditions for work: 1. A force must act on the object. 2. The object must undergo displacement in the direction of the force.

Keywords: force, displacement, direction of force

2. Zero Work

Answer: Work done is zero in the following cases: 1. When a person holds a heavy load but does not move it, displacement is zero. 2. When a person walks carrying a load on a horizontal road, the force applied is vertical while displacement is horizontal. 3. In circular motion, centripetal force is perpendicular to displacement, so work done is zero.

Keywords: zero displacement, perpendicular force, circular motion

3. Positive, Negative and Zero Work

Answer: - **Positive work:** When force and displacement are in the same direction (e.g., gravity on a falling stone). - **Negative work:** When force acts opposite to displacement (e.g., friction on a moving object). - **Zero work:** When force is perpendicular to displacement or displacement is zero.

Keywords: same direction, opposite direction, perpendicular

4. Kinetic Energy

Answer: Kinetic energy is the energy possessed by a body due to its motion.

Derivation: Let a body of mass m move with velocity v . Work done = Change in kinetic energy Using equations of motion and Newton's laws, Kinetic Energy, $E_k = \frac{1}{2}mv^2$

Keywords: motion, work-energy theorem, $\frac{1}{2}mv^2$

5. Potential Energy

Answer: Gravitational potential energy is the energy possessed by a body due to its position or height above the ground.

Derivation: Work done against gravity in lifting an object of mass m through height h : $PE = mgh$ It depends only on height and not on the path because gravity is a conservative force.

Keywords: height, gravity, conservative force

6. Law of Conservation of Energy

Answer: Energy can neither be created nor destroyed; it can only be transformed from one form to another.

For a freely falling body: At top: PE maximum, KE zero During fall: PE decreases, KE increases At ground: KE maximum, PE zero Total energy remains constant.

Keywords: transformation, total energy constant, free fall

7. Power

Answer: Power is the rate of doing work. Power = Work / Time

SI unit: Watt (W) 1 Watt is the power when 1 joule of work is done in 1 second.

Keywords: rate of work, watt, joule per second

8. Commercial Unit of Energy

Answer: The commercial unit of electrical energy is kilowatt-hour (kWh).

$1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ s} = 3.6 \times 10^6 \text{ J}$

Keywords: electrical energy, unit, joule

9. Energy Transformations

Answer: (a) Electric bulb: Electrical $\rightarrow$ Light + Heat (b) Solar cell: Solar $\rightarrow$ Electrical (c) Battery: Chemical $\rightarrow$ Electrical (d) Microphone: Sound $\rightarrow$ Electrical

Keywords: transformation, input energy, output energy

10. Mechanical Energy

Answer: Mechanical energy is the sum of kinetic and potential energy. In an ideal system, mechanical energy remains constant due to conservation of energy.

Keywords: KE + PE, ideal system, conservation

11. Work Done by Gravity

Answer: While carrying a suitcase horizontally, displacement is horizontal but gravitational force acts vertically downward. Hence, work done by gravity is zero.

Keywords: perpendicular, gravity, zero work

12. Stretched Bow

Answer: Elastic potential energy stored in the stretched bow is converted into kinetic energy of the arrow.

Keywords: elastic potential energy, kinetic energy

13. Energy at Rest

Answer: Yes, an object at rest can possess energy. For example, a stone kept at a height has gravitational potential energy.

Keywords: position, stored energy, potential

14. Power vs Energy

Answer: Both girls do the same work as they climb the same height. Girl A has more power because she takes less time.

Keywords: same work, time, rate

15. Work Calculation (Numerical)

Given: $F = 10 \text{ N}$, $s = 2 \text{ m}$ Work = $Fs = 10 \times 2 = 20 \text{ J}$

Keywords: $W = Fs$, joule

16. Change in Kinetic Energy

Answer: Convert speeds to m/s. Use $KE = \frac{1}{2}mv^2$ Work required = Change in KE

Keywords: change in KE, work

17. Potential Energy (Numerical)

$PE = mgh = 10 \times 9.8 \times 6 = 588 \text{ J}$

Keywords: height, gravity, PE

18. Power of Motor

Answer: Work = Power $\times$ Time = $2000 \times 60 = 1.2 \times 10^5 \text{ J}$

Keywords: power, time, work

19. Household Consumption

Answer: Energy = $250 \times 3.6 \times 10^6 \text{ J} = 9 \times 10^8 \text{ J}$

Keywords: unit conversion, kWh

20. Device Rating

Answer: Energy = Power $\times$ Time = $100 \text{ W} \times 8 \text{ h} = 0.8 \text{ kWh}$

Keywords: watt, unit, consumption

✓ This document is fully exam-ready and aligned with CBSE 2026 marking scheme.